1. Monday, June 17. Crash course on set Theory, functions, relations.

(Naive) Set Theory:

Definition: A set is a well-defined collection of objects.

Say a is an element of a set A by writing $a \in A$. Write $a \notin A$ if a is not an element of A.

We use $A \subseteq B$ to say that A is a subset of B (formally: $\forall x \in A,\ x \in A \Rightarrow x \in B$)

Basic set operations

$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$.

Facts There is an empty set $\emptyset$ with no elements.

Associative: $A \cup (B \cup C) = (A \cup B) \cup C$ so makes sense to write $A \cup B \cup C$.

Distributive: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

De Morgan's rules: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$

$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

Families of sets: Let $\mathcal{A}$ be a nonempty collection of sets.

Then $\bigcup_{A \in \mathcal{A}} A = \{ x \mid \text{There exists } A \in \mathcal{A} \text{ such that } x \in A \}$

$\bigcap_{A \in \mathcal{A}} A = \{ x \mid x \in A \text{ for all } A \in \mathcal{A} \}$.

Distributivity and De'Morgan's hold for families.

In practice, $\mathcal{A}$ is indexed: There is a set Λ such that

$$\mathcal{A} = \{ A_\alpha \mid \alpha \in \Lambda \}$$

We denote $\bigcup_{A \in \mathcal{A}} A = \bigcup_{i \in \Lambda} A_i$ and also $\bigcap$

Cartesian product:

$$A \times B = \{ (a,b) \mid a \in A, b \in B \}$$

Here $(a,b) = (a',b') \iff a = a', \; b = b'$.

Functions: A function $f: A \to B$ is a "rule of assignment from A to B"

Formally, a function from A to B is a subset $f \subseteq A \times B$ such that:

1) $\forall a \in A \ \exists\, b \in B$ such that $(a,b) \in f$.
2) If $(a,b_1) \in f$ and $(a,b_2) \in f$, Then $b_1 = b_2$.

So, for each $a \in A$ there is a <u>unique</u> $b \in B$ with $(a,b) \in f$.

Makes sense to write $b = f(a)$!!

Notation: If $U \subseteq A$, denote $f(U) = \{ f(a) \mid a \in U \} \subseteq B$

If $V \subseteq B$, denote $f^{-1}(V) = \{ a \in A \mid f(a) \in V \} \subseteq A$.

Q: $f(U \cup V) \overset{?}{=} f(U) \cup f(V)$? $\qquad$ $f(U \cap V) \overset{?}{=} f(U) \cap f(U)$?

$f^{-1}(U \cup V) \overset{?}{=} f^{-1}(U) \cup f^{-1}(V)$? $\qquad$ $f^{-1}(U \cap V) \overset{?}{=} f^{-1}(U) \cap f^{-1}(U)$?

Upshot: f^{-1} behaves better.

A function is injective if $f(a) = f(a') \Rightarrow a = a'$

A function $f: A \to B$ is surjective if $f(A) = B$ (i.e. $\forall b \in B \; \exists a \in A$ with $f(a) = b$)

A function is bijective if inj. and surj.

Composition: Given $f: A \to B$ and $g: B \to C$, have $g \circ f: A \to C$, $(g \circ f)(a) = g(f(a))$

Notation: $A \xrightarrow{f} B \xrightarrow{g} C$

$a \longmapsto f(a) \longmapsto g(f(a))$

Composition is associative.

For any set A, we have the identity function $I_A: A \to A$, $a \mapsto a$

We have $g \circ I_A = g$, $I_B \circ f = f$.

Fact: A function $f: A \to B$ is bijective $\Leftrightarrow \exists$ function $f^{-1}: B \to A$ such that

$f^{-1} \circ f = I_A$ and $f \circ f^{-1} = I_B$.

Cardinality: A set A is finite if $\exists\ k \in \mathbb{N}$ and bijection $A \longrightarrow \{1, \ldots, k\}$.

We also include $\emptyset$ here.

Definition:
- A set is **infinite** if it is not finite.
- **Countably infinite** if $\exists\ A \longrightarrow \mathbb{N}$ bijective. $\mathbb{Z}$, $\mathbb{Q}$
- **Countable** if finite or countable infinite.
- **Uncountable** if not countable $\mathbb{R}$, $\mathcal{P}(\mathbb{Z})$

Facts: 1) Subset of countable is countable.

2) Countable union of countable sets is countable.

3) Finite product of countable sets is countable.

Equivalence relations.

Definition: A relation on a set A is a subset $R \subseteq A \times A$.

We write aRb (or $a \sim b$ when R is clear) instead of $(a,b) \in R$.

We say R is:

- **Reflexive** if $aRa \quad \forall a \in A$.
- **Symmetric** if $aRb \Rightarrow bRa$.
- **Transitive** if aRb and $bRc \Rightarrow aRc$.

We say R is an **eq. Relation** if all three hold.

Definition: If R is an equivalence Relation on A, given $a \in A$ we define the **equivalence class** of a as

$$[a] = \{ x \in A \mid xRa \}.$$

Theorem: If R is an equiv. relation on A, Then equivalence classes partition A.

i.e. each $a \in A$ belongs to a unique equivalence class.